Growth and boundedness at the critical Chambolle–Pock endpoint

Abstract

At extrapolation $1/8$ and step product $r_* = (88+16\sqrt{91})/81$, the literal primal-first Chambolle–Pock method has bounded increments but need not have bounded positions. We give a coherent account of this critical dynamics. An exact semicoercive certificate bounds the increments of every initialized Hilbert-space orbit. A signed storage then proves an explicit $O(\sqrt{N})$ maximal-radius estimate whenever a KKT pair exists; the storage theorem itself holds at all positive steps and every real extrapolation parameter under bounded increments. A single explicit endpoint orbit has compact dual set, unique KKT pair, and primal norm at least $\sqrt{n}/[16(1+\log_4 n)]$ at every positive time. The construction establishes sharpness only of the universal polynomial exponent. Bounded dual domain together with primal-objective coercivity restores boundedness at all positive steps and nonnegative extrapolation, with quantitative interior-ball and finite-dimensional consequences. We also retain boundedness for eventual periodic affine itineraries and a no-drift obstruction for arbitrary monotone graphs. These results separate unique solvability, coercivity, and finite-dimensional structure; general nonseparable finite-dimensional endpoint boundedness remains open.

Keywords: Chambolle–Pock iteration; critical endpoint; square-root growth; Hilbert-space escape; coercive boundedness; signed storage

2020 Mathematics Subject Classification. Primary 39A22; Secondary 47J25, 90C25, 65K10.

1 The critical dynamics question

Let X, Y be real Hilbert spaces, let $L : X \to Y$ be bounded and linear, and let $f : X \to (-\infty, +\infty]$ and $h : Y \to (-\infty, +\infty]$ be proper lower-semicontinuous convex functions. Fix $\tau, \sigma > 0$ and $\theta \in \mathbb{R}$. From any initialized pair $(x_0, y_0) \in X \times Y$, consider exactly

$$\begin{aligned} x_n &= \operatorname{prox}_{\tau f}(x_{n-1} - \tau L^* y_{n-1}), \\ y_n &= \operatorname{prox}_{\sigma h}\big(y_{n-1} + \sigma L(x_n + \theta(x_n - x_{n-1}))\big), \qquad n \geq 1. \end{aligned} \tag{1}$$

Here $\operatorname{prox}_{\tau f}(w)$ minimizes $f(x) + \|x - w\|^2/(2\tau)$. The order and the extrapolated point in (1) are part of every assertion. We assume a KKT pair (x_*, y_*) exists when expressly stated:

$$-L^* y_* \in \partial f(x_*), \qquad L x_* \in \partial h(y_*). \tag{2}$$

The function h is the dual-side function used in the iteration; it may be written as a conjugate in saddle-point notation. We do not require bounded proximal domains, coercivity, strong convexity, or finite dimension for the principal growth theorem.

The critical product is

$$r_* = \frac{88 + 16\sqrt{91}}{81}, \qquad 81r_*^2 - 176r_* - 192 = 0.$$

As a prior prerequisite, report [4, Theorem 1] proves that at $\theta = 1/8$ every KKT-anchored orbit of (1) with $\tau\sigma \|L\|^2 < r_*$ is bounded, has square-summable increments and vanishing KKT residuals, and converges weakly to a KKT pair; convergence is in norm in finite dimensions. Its nonconstant planar three-cycle at equality makes that universal strict threshold sharp. We do not re-count that theorem as a result of this manuscript. The unchanged prerequisite PDF, source and exact certificate are supplied with this archive. The endpoint increment argument below reconstructs its needed finite identities explicitly, rather than importing the strict proof or a private computation.

The question at equality is therefore how far an actual orbit can escape, and which additional assumptions exclude escape. The answer is not supplied by uniqueness of the KKT pair: in infinite dimension even a compact dual set and a unique pair permit unbounded primal growth. The positive position bound, the matching polynomial-exponent obstruction, and the sufficient boundedness mechanisms are developed with the same map and KKT convention throughout.

Theorem 1 (Square-root growth from bounded increments)**.** *Assume* (2) *and uniformly bounded successive increments in* (1)*, with any fixed* $\theta \in \mathbb{R}$*. Define*

$$u_n = \frac{x_n - x_*}{\sqrt{\tau}}, \qquad v_n = \frac{y_n - y_*}{\sqrt{\sigma}}, \qquad K = \sqrt{\tau\sigma}\, L, \qquad a = \|K\|,$$

and let $\rho_n = (\|u_n\|^2 + \|v_n\|^2)^{1/2}$*. Choose a finite* $d \geq 0$ *with* $\|u_n - u_{n-1}\|, \|v_n - v_{n-1}\| \leq d$ *for all* $n \geq 1$*. Put*

$$e = d\big(1 + a\max\{1, |\theta|\}\big), \qquad \eta = e(1 + 2|1 - \theta|),$$
$$B_\theta = \begin{cases} 2d^2\big[(1-\theta)(1+a) + |\theta|a\big], & \theta \leq 1, \\ 2\theta a d^2, & \theta > 1. \end{cases} \tag{3}$$

Then, for every $n \geq 1$*,*

$$\rho_n \leq \eta + \sqrt{(\rho_1 + \eta)^2 + B_\theta(n-1)}.$$

Consequently, for every integer $N \geq 1$*,*

$$\max_{0 \leq n \leq N} \rho_n \leq \max\left\{\rho_0,\ \eta + \sqrt{(\rho_1 + \eta)^2 + B_\theta(N-1)}\right\} = O(\sqrt{N}). \tag{4}$$

If $0 \leq \theta \leq 1$*, both bounds improve by replacing* η *with* $e = d(1 + a)$*; in that range* $B_\theta = 2d^2(1 - \theta + a)$*. No finite function value at initialization is assumed. In particular,* $x_n/n \to 0$ *and* $y_n/n \to 0$ *in norm.*

The increment bound is a hypothesis at unrestricted parameters; it is not asserted for all orbits of (1). At the specific endpoint

$$\theta = \frac{1}{8}, \qquad \tau\sigma \left\|L\right\|^2 = r_*, \qquad r_* = \frac{88 + 16\sqrt{91}}{81}, \tag{5}$$

Section 3 supplies it without a KKT assumption. Hence every KKT-anchored endpoint orbit satisfies the square-root bound. Section 4 gives the nearly square-root lower example. Section 5 establishes boundedness under primal-objective coercivity and explains the finite-dimensional contrast. Section 6 supplies quantitative interior-ball bounds. Section 7.1 gives a conditional finite-dimensional periodic affine-itinerary bound. Section 7.2 proves an obstruction using only monotone graph pairs.

The Chambolle–Pock method originates in [1]. Our results concern necessary restrictions on escape in the literal order (1); they do not invoke a general convergence theorem for this map at these parameters. In particular, no nonexpansive metric for the full endpoint map is assumed. We make no exhaustive priority claim for the elementary graph arguments or the growth estimate. For context, the sufficient small-extrapolation bound in [6, Theorem 1.2] specializes to 120/113 at $\theta = 1/8$. The bound $4/(1 + 2\theta)$ in [7, Theorem 2] assumes $\theta > 1/2$ and is not a sufficiency theorem at 1/8; its scalar bilinear example gives the necessary ceiling $16/5 > r_*$. These comparisons preserve their stated source versions and parameter scopes.

2 A signed storage gives square root growth

2.1 Normalize at one fixed KKT pair

Define

$$\begin{aligned} F(u) &= f(x_* + \sqrt{\tau}u) - f(x_*) + \sqrt{\tau}\left\langle L^* y_*, u\right\rangle, \\ G(v) &= h(y_* + \sqrt{\sigma}v) - h(y_*) - \sqrt{\sigma}\left\langle Lx_*, v\right\rangle. \end{aligned} \tag{6}$$

The KKT inclusions give $F(0) = G(0) = 0$ and $F, G \geq 0$. Both functions are proper lower-semicontinuous convex. Put $\Delta u_n = u_n - u_{n-1}$ and $\Delta v_n = v_n - v_{n-1}$. The actual proximal inclusions in (1) are

$$\begin{aligned} p_n &= -\Delta u_n - K^* v_{n-1} \in \partial F(u_n), \\ q_n &= -\Delta v_n + K(u_n + \theta \Delta u_n) \in \partial G(v_n). \end{aligned} \tag{7}$$

Every $F_n := F(u_n)$ and $G_n := G(v_n)$ at $n \geq 1$ is finite. No function value or subgradient is required at the arbitrary initialization.

Introduce

$$\begin{aligned} \alpha_n &= -\Delta u_n + K^* \Delta v_n, \quad \beta_n = -\Delta v_n + \theta K \Delta u_n, \\ C_n &= \left\langle Ku_n, v_n\right\rangle, \qquad\qquad E_n = \left\|u_n\right\|^2 + \left\|v_n\right\|^2 = \rho_n^2. \end{aligned} \tag{8}$$

Then $p_n = -K^* v_n + \alpha_n$ and $q_n = K u_n + \beta_n$. For arbitrary real θ, the increment bound gives

$$\|\alpha_n\|, \|\beta_n\| \le e = d(1 + a \max\{1, |\theta|\}). \tag{9}$$

Convexity tested at the common minimizing anchor yields

$$0 \le F_n \le -C_n + \langle \alpha_n, u_n \rangle, \qquad 0 \le G_n \le C_n + \langle \beta_n, v_n \rangle. \tag{10}$$

2.2 A quadratic lower bound with a linear defect

For $n \ge 1$, define the real signed storage

$$W_n = E_n - 2C_n - 2(1-\theta)F_n. \tag{11}$$

It need not be positive on arbitrary states or histories. Along this actual KKT-anchored orbit, (10) gives

$$|C_n| \le e\rho_n, \qquad 0 \le F_n \le 2e\rho_n.$$

Therefore the constant η in (3) satisfies

$$|W_n - E_n| \le 2|C_n| + 2|1-\theta| F_n \le 2\eta\rho_n. \tag{12}$$

In particular, $W_n \ge \rho_n^2 - 2\eta\rho_n$ and $W_1 \le \rho_1^2 + 2\eta\rho_1$.

For the sharper constants on $0 \le \theta \le 1$, keep the signs in the anchor inequalities. Since $\theta F_n \ge 0$,

$$\begin{aligned} W_n &= E_n - 2(C_n + F_n) + 2\theta F_n \\ &\ge E_n - 2\langle \alpha_n, u_n \rangle \ge \rho_n^2 - 2e\rho_n. \end{aligned} \tag{13}$$

Also $1 - \theta \ge 0$, so

$$W_1 \le E_1 - 2C_1 \le \rho_1^2 + 2\langle \beta_1, v_1 \rangle \le \rho_1^2 + 2e\rho_1. \tag{14}$$

These two one-sided bounds permit replacing η by e in that range. No positive-definite metric is presumed.

2.3 The exact coupling product rule

Test (7) at zero and add:

$$F_n + G_n \le -\langle \Delta u_n, u_n \rangle - \langle \Delta v_n, v_n \rangle + \langle K u_n, \Delta v_n \rangle + \theta \langle K \Delta u_n, v_n \rangle.$$

Since $F_n + G_n \ge 0$, polarization yields

$$\begin{aligned} E_n - E_{n-1} \le & -\|\Delta u_n\|^2 - \|\Delta v_n\|^2 \\ & + 2\langle K u_n, \Delta v_n \rangle + 2\theta \langle K \Delta u_n, v_n \rangle. \end{aligned} \tag{15}$$

For the temporary coupling terms

$$\mathcal{B}_n = \langle K\Delta u_n, v_{n-1}\rangle\,, \qquad H_n = \langle K\Delta u_n, \Delta v_n\rangle\,,$$

the exact identities are

$$C_n - C_{n-1} = \langle Ku_n, \Delta v_n\rangle + \mathcal{B}_n, \qquad \langle K\Delta u_n, v_n\rangle = \mathcal{B}_n + H_n.$$

Therefore (15) becomes

$$\begin{aligned} E_n - E_{n-1} \le\ & -\|\Delta u_n\|^2 - \|\Delta v_n\|^2 \\ & + 2(C_n - C_{n-1}) + 2(\theta-1)\mathcal{B}_n + 2\theta H_n. \end{aligned} \tag{16}$$

2.4 Adjacent output convexity telescopes the remaining term

First suppose $\theta \le 1$, and start at $n = 2$. Test the actual subgradient p_{n-1} at u_{n-1} against u_n:

$$F_n - F_{n-1} \ge -\langle \Delta u_{n-1}, \Delta u_n\rangle - \langle K\Delta u_n, v_{n-2}\rangle\,.$$

Using $v_{n-1} = v_{n-2} + \Delta v_{n-1}$ gives

$$\mathcal{B}_n \ge F_{n-1} - F_n - \langle \Delta u_{n-1}, \Delta u_n\rangle + \langle K\Delta u_n, \Delta v_{n-1}\rangle\,. \tag{17}$$

The coefficient $2(\theta-1)$ in (16) is nonpositive, so multiplying (17) by it reverses the inequality. Move the function-value and C_n differences to the left. The result is precisely

$$\begin{aligned} W_n - W_{n-1} \le\ & -\|\Delta u_n\|^2 - \|\Delta v_n\|^2 \\ & + 2(1-\theta)\,\langle \Delta u_{n-1}, \Delta u_n\rangle \\ & - 2(1-\theta)\,\langle K\Delta u_n, \Delta v_{n-1}\rangle + 2\theta\,\langle K\Delta u_n, \Delta v_n\rangle\,. \end{aligned} \tag{18}$$

Drop the first two nonpositive terms, use $1-\theta \ge 0$, and bound the final mixed term by its absolute value. This yields

$$\begin{aligned} W_n - W_{n-1} &\le 2(1-\theta)d^2 + 2(1-\theta)ad^2 + 2|\theta|ad^2 \\ &= 2d^2\big[(1-\theta)(1+a) + |\theta|a\big] = B_\theta. \end{aligned} \tag{19}$$

For $0 \le \theta \le 1$ this reduces to $B_\theta = 2d^2(1-\theta+a)$.

If $\theta > 1$, use the current primal subgradient p_n tested at the previous output instead. For $n \ge 2$,

$$F_{n-1} - F_n \ge \|\Delta u_n\|^2 + \langle K\Delta u_n, v_{n-1}\rangle\,.$$

This is an upper bound on $\mathcal{B}_n$. Its coefficient $2(\theta-1)$ in (16) is now positive. Substitution and the definition of W_n give

$$\begin{aligned} W_n - W_{n-1} &\le -(2\theta-1)\,\|\Delta u_n\|^2 - \|\Delta v_n\|^2 + 2\theta\,\langle K\Delta u_n, \Delta v_n\rangle \\ &\le 2\theta a d^2 = B_\theta. \end{aligned} \tag{20}$$

In both branches the fixed B_θ is nonnegative, and only positive-index function values are used. The temporary coupling $\mathcal{B}_n$ has been eliminated.

2.5 Sum and solve the scalar inequality

Summing the appropriate branch from two through n gives, also trivially for $n = 1$,

$$W_n \le W_1 + B_\theta(n-1).$$

Using (12) at n and at one, we obtain

$$\rho_n^2 - 2\eta\rho_n \le \rho_1^2 + 2\eta\rho_1 + B_\theta(n-1).$$

Completing the square yields

$$(\rho_n - \eta)^2 \le (\rho_1 + \eta)^2 + B_\theta(n-1).$$

The right side is nonnegative. Taking its positive square root proves the individual bound in Theorem 1. Because $B_\theta \ge 0$, the right side is nondecreasing in n. Taking the maximum and adding the separately finite ρ_0 proves (4).

When $0 \le \theta \le 1$, repeat this last argument with the sharper one-sided bounds (13)–(14); the same calculation replaces η with e. Dividing by n also gives $x_n/n, y_n/n \to 0$ in norm. All arguments are Hilbert inner-product identities or convex subgradient inequalities; no finite-dimensional compactness is involved. This completes the proof of Theorem 1.

3 Bounded increments at the one eighth endpoint

The next fact is independent of the existence of a KKT pair. We state its exact finite certificate and give the full passage to a Hilbert-space orbit. The source archive contains the unchanged serialized coefficient file and a verifier using only Python's standard library.

Proposition 3.1 (Endpoint increments)**.** For proper lower-semicontinuous convex f, h on real Hilbert spaces and bounded linear L, every initialized orbit of (1) satisfying (5) has uniformly bounded successive increments. A KKT pair is not required.

3.1 The exact finite certificate

All coefficients belong to $\mathbb{Q}(\sqrt{91})$. The public file `exact_certificate.json` stores the parametric family from companion report [4]; here only its constant coefficient at the endpoint is used. An entry $[a, b]$ represents $a + b\sqrt{91}$ with rational a, b. Let P_0 be the symmetric 8×8 matrix whose 36 upper-triangular entries are the first array under `P_numerator`, in lexicographic order (i, j), $0 \le i \le j < 8$. Let λ and μ be the first arrays under their respective keys, and c_0 the first array under `c`. We use the numerator storage $\widehat{P} = P_0$; this is r_* times the unscaled endpoint storage in the parametric family.

Introduce ten scalar formal slots $w = (X_0, Y_0, \dots, X_4, Y_4)$ and coordinate selectors $e_{i,s}$, where $s = 0$ denotes a primal slot and $s = 1$ a dual slot. For $i \geq 1$, define the formal gradients

$$g_{i,0} = \frac{e_{i-1,0} - e_{i,0}}{r_*} - e_{i-1,1}, \qquad g_{i,1} = e_{i-1,1} - e_{i,1} + \frac{9e_{i,0} - e_{i-1,0}}{8}. \tag{21}$$

For an edge $e = (i, j, s)$, its formal convexity slack is

$$S_e = \phi_{j,s} - \phi_{i,s} - \langle g_{i,s}, z_{j,s} - z_{i,s} \rangle = w^{\mathsf{T}} Q_e w + v_e^{\mathsf{T}} \phi. \tag{22}$$

The optional anchor index $\varnothing$ has zero point, value, and gradient; there is no gradient at index zero. Within each $s = 0, 1$, order the descent edges by

$$(\varnothing, k)\ (0 \leq k \leq 4), \qquad (k, \varnothing)\ (1 \leq k \leq 4), \qquad (i, j)\ (1 \leq i \leq 4,\ 0 \leq j \leq 4,\ i \neq j),$$

with the last list lexicographic. There are 50 edges. The 32 lower-bound edges are this list with every edge involving index four removed. The coefficients λ_e, μ_e follow these orders.

Let O select the first eight slots and N the last eight. The exact data satisfy

$$\begin{aligned} \widehat{D} &= O^{\mathsf{T}} \widehat{P} O - N^{\mathsf{T}} \widehat{P} N - r_* \sum_e \lambda_e Q_e \succeq 0, \\ O^{\mathsf{T}} \widehat{C}_8 O &= O^{\mathsf{T}} \widehat{P} O - r_* \sum_e \mu_e Q_e \succeq 0, \\ c_0 &= 0, \qquad \sum_e \lambda_e v_e = 0, \qquad \sum_e \mu_e v_e = 0. \end{aligned} \tag{23}$$

All λ_e, μ_e are nonnegative. Every nonzero coefficient uses ordinary positive indices $i, j \geq 1$, so there is no nonzero anchor edge or index-zero edge. The matrix $\widehat{P}$ annihilates the two constant-history columns

$$Z_8 = [I_2; I_2; I_2; I_2].$$

The kernel of $\widehat{C}_8$ is exactly $\operatorname{ran} Z_8$.

Here is a finite exact way to check the positivity and kernel assertions rather than relying on numerical eigenvalues. For $\widehat{C}_8$, take the first six coordinate columns E_6 and Z_8. The checker verifies that $[E_6, Z_8]$ is invertible, that $\widehat{C}_8 Z_8 = 0$, and that $E_6^{\mathsf{T}} \widehat{C}_8 E_6$ has a strictly positive exact LDL^{T} diagonal. For $\widehat{D}$, take the first four coordinate columns E_4 and six kernel columns comprising constants, two ramps, and two cycle columns. The constant and ramp rows at pair i are I_2 and iI_2. To specify the cycle columns, repeat

$$(c_i, s_i) = (1, 0), \quad (-1/2, 1/2), \quad (-1/2, -1/2)$$

with period three, and use the two rows

$$(c_i, s_i), \qquad \left(\frac{9}{16} c_i + \frac{3 s_i}{2 r_*},\ \frac{9}{16} s_i - \frac{c_i}{2 r_*} \right) \qquad (0 \leq i \leq 4).$$

With these six columns denoted by H, the checker verifies $\widehat{D}H = 0$, invertibility of $[E_4, H]$, and strict positivity of the four exact LDL^{T} pivots of $E_4^{\mathsf{T}}\widehat{D}E_4$. Congruence therefore proves both positive-semidefiniteness statements and their exact kernel dimensions, two and six respectively. Rational sign tests in $\mathbb{Q}(\sqrt{91})$ are performed by comparing rational squares when the two terms have opposite signs.

The certificate file has SHA-256

```
421ea0abb9e4bb550d19310aa0c85d90
db54531b181c1378796904bff5b97815
```

The two printed lines are concatenated to form the hash. The endpoint checker pins this exact file, reconstructs every slack and value cancellation in (23), and checks the complementary blocks and kernel bases. The complete rational pivots are regenerated in a JSON result file; the checker never trusts a pre-existing result file.

3.2 Transfer the certificate to the actual orbit

Normalize without any solution anchor:

$$X_n = \sqrt{r_*/\tau}\, x_n, \quad Y_n = y_n/\sqrt{\sigma}, \quad K_0 = \sqrt{\tau\sigma/r_*}\, L, \quad \|K_0\| = 1. \tag{24}$$

Put $\mathcal{F}(X) = f(\sqrt{\tau/r_*}\, X)$ and $\mathcal{H}(Y) = h(\sqrt{\sigma}Y)$. Direct substitution into (1) gives

$$\begin{aligned} (X_{n-1} - X_n)/r_* - K_0^* Y_{n-1} &\in \partial\mathcal{F}(X_n), \\ Y_{n-1} - Y_n + K_0(9X_n - X_{n-1})/8 &\in \partial\mathcal{H}(Y_n). \end{aligned} \tag{25}$$

For a contraction K_0, the maps into $Y \oplus X$

$$JX = (K_0 X, (I - K_0^* K_0)^{1/2} X), \qquad I_Y Y = (Y, 0)$$

are isometries on their respective spaces and satisfy

$$\langle JX, I_Y Y \rangle = \langle K_0 X, Y \rangle .$$

Thus they preserve every same-variable pairing and represent every cross pairing in the formal quadratic slacks. This transfers the scalar matrix identities to the Hilbert-valued histories. In particular, each weighted formal slack becomes exactly an actual subgradient convexity slack from (25). No subgradient on the embedding space is posited.

For $k \geq 1$, let

$$h_k = (JX_k, I_Y Y_k, \ldots, JX_{k+3}, I_Y Y_{k+3}), \qquad \mathcal{E}_k = \left\langle \widehat{P} h_k, h_k \right\rangle,$$

and let w_k be the corresponding five-pair history. The value cancellations and positive-index support in (23) give

$$\begin{aligned} \mathcal{E}_k &= \left\langle \widehat{C}_8 h_k, h_k \right\rangle + r_* \sum_e \mu_e S_{e,k} \geq 0, \\ \mathcal{E}_k - \mathcal{E}_{k+1} &= \left\langle \widehat{D} w_k, w_k \right\rangle + r_* \sum_e \lambda_e S_{e,k} \geq 0. \end{aligned} \tag{26}$$

All values appearing in a nonzero slack are at proximal outputs and are finite. There is no use of a value at the initialized point or a KKT anchor. Hence $0 \le \mathcal{E}_k \le \mathcal{E}_1$ for $k \ge 1$.

Let $\kappa > 0$ be the smallest eigenvalue of $\widehat{C}_8$ on the orthogonal complement of its constant-history kernel. Orthogonal diagonalization of this finite scalar matrix works for Hilbert-valued slots. If $\bar{U}, \bar{V}$ are the separate means of the four primal and dual slots, it follows that

$$\left\langle \widehat{C}_8 h, h \right\rangle \ge \kappa \sum_{i=0}^{3} (\left\|U_i - \bar{U}\right\|^2 + \left\|V_i - \bar{V}\right\|^2).$$

Since $\|a-b\|^2 \le 2\|a\|^2 + 2\|b\|^2$ and both embeddings are isometries, (26) implies

$$\mathcal{E}_k \ge \frac{\kappa}{2}\big(\|X_{k+3} - X_{k+2}\|^2 + \|Y_{k+3} - Y_{k+2}\|^2\big). \tag{27}$$

This bounds all tail increments. The finite initial prefix consists of finite Hilbert-space points. Undoing the fixed positive scalings proves Proposition 3.1.

Neither $\widehat{P}$ nor this argument provides a coercive position bound. In particular, $\widehat{C}_8$ has constant histories in its kernel, and $\widehat{D}$ additionally has ramp and cycle directions in its kernel. It would be invalid to import the strict coercivity of a below-endpoint certificate by continuity at equality.

Corollary 3.1 (Square-root endpoint growth)**.** *Under* (5) *and* (2)*, every initialized orbit satisfies*

$$\max_{0 \le n \le N} \left(\frac{\|x_n - x_*\|^2}{\tau} + \frac{\|y_n - y_*\|^2}{\sigma} \right)^{1/2} = O(\sqrt{N}).$$

The sharper $0 \le \theta \le 1$ *version of* (4) *applies with any uniform normalized increment bound* d *supplied by Proposition 3.1, replacing* η *by* e*, where*

$$e = d(1 + \sqrt{r_*}), \qquad B_{1/8} = d^2(7/4 + 2\sqrt{r_*}).$$

Proof. The proposition supplies precisely the remaining hypothesis of Theorem 1. The normalizations in (24) and the theorem differ only by fixed positive scalings and translations. Substituting $a = \sqrt{r_*}$ and $\theta = 1/8$ gives the displayed constants. ∎

4 An endpoint orbit with nearly square root escape

The upper estimate does not imply boundedness in Hilbert space. A single cylinder–tetrahedron construction supplies the lower examples. Its strongest rescaling gives one orbit that grows faster than every power n^α with $\alpha < 1/2$. The entire construction and its proof are given here.

Theorem 2 (A nearly square-root lower bound)**.** *There exist a separable real Hilbert space E and nonempty closed convex sets $C, D \subset E$ such that D is norm compact, their identity-coupled KKT set is exactly $\{(0,0)\}$, and an initialized actual orbit*

$$x_{n+1} = P_C(x_n - r_* y_n), \qquad y_{n+1} = P_D\left(y_n + \frac{9x_{n+1} - x_n}{8}\right) \tag{28}$$

satisfies, for every integer $n \geq 1$,

$$\|x_n\| \geq \frac{\sqrt{n}}{16(1+\log_4 n)}. \tag{29}$$

In particular, $\|x_n\|/n^\alpha \to +\infty$ for every $\alpha < 1/2$. The dual orbit is bounded. The parameters are $\tau = r_*$, $\sigma = 1$, $L = I_E$, and $\theta = 1/8$.

4.1 Exact planar data and the finite drifting block

Write $r = r_*$ and $q = \sqrt{91}$. Equip $\mathbb{R}^2$ with the inner product $\langle u, v\rangle_M = u_1 v_1 + 3u_2 v_2$ and put

$$R = \begin{pmatrix} -1/2 & -3/2 \\ 1/2 & -1/2 \end{pmatrix}, \qquad c_i = R^i \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad d_i = R^i \begin{pmatrix} 9/16 \\ -1/(2r) \end{pmatrix}. \tag{30}$$

All phase indices are taken modulo three. Direct multiplication gives $R^3 = I$ and $R^{\mathsf{T}} M R = M$, where $M = \operatorname{diag}(1,3)$. Both triples sum to zero. For every nonzero planar vector v,

$$\det(v, Rv) = \frac{v_1^2 + 3v_2^2}{2} > 0, \qquad \det(Rv - v, R^2 v - v) = 3\det(v, Rv).$$

Both triples therefore span nondegenerate triangles with zero in their interiors.

Define

$$p_i = c_{i-1} - r d_{i-1} - c_i, \qquad w_i = d_{i-1} + (9c_i - c_{i-1})/8 - d_i, \tag{31}$$

and set

$$m = \frac{q-8}{18}, \qquad s = \frac{10q - 91}{128}, \qquad \gamma = rs = \frac{91 - 8q}{144}. \tag{32}$$

These constants are positive and

$$\gamma > \frac{11}{144} > \frac{9}{256}.$$

Indeed $q < 10$ gives the first strict bound, and $q > 91/10$ gives $s > 0$, both by rational square comparisons. The phase-zero residuals and vertex tests are exactly

$$\begin{aligned} p_0 &= (m, m), & w_0 &= ((11-q)/16, (q-9)/16), \\ \big(\langle p_0, c_k - c_0\rangle_M\big)_{k=0}^2 &= (0, 0, -3m), & \big(\langle w_0, d_k - d_0\rangle_M\big)_{k=0}^2 &= (0, 0, -3s), \\ \langle w_0, d_0\rangle_M &= s. \end{aligned} \tag{33}$$

Rotation gives the corresponding inequalities and $\langle w_i, d_i \rangle_M = s$ at every phase. These are identities in $\mathbb{Q}(\sqrt{91})$.

Let $E_0 = \mathbb{R}^2 \times \mathbb{R}$ with inner product $\mathrm{diag}(1, 3, 1)$. For $\varepsilon > 0$, define

$$\begin{aligned} C^\varepsilon &= \mathrm{conv}\{c_0, c_1, c_2\} \times \mathbb{R}, \\ D^\varepsilon &= \mathrm{conv}\{v^+, v_0^-, v_1^-, v_2^-\}, \qquad v^+ = (0, \varepsilon/r), \quad v_i^- = (d_i, -\varepsilon/r). \end{aligned} \tag{34}$$

The first zero in v^+ is a planar vector. Initialize (28) on these sets at $x_0 = (c_0, 0)$, $y_0 = (d_0, -\varepsilon/r)$. We claim that, whenever

$$2(N + 1/8)\varepsilon^2 \le \gamma, \tag{35}$$

the actual orbit is

$$x_n = (c_{n \bmod 3}, n\varepsilon), \qquad y_n = (d_{n \bmod 3}, -\varepsilon/r), \qquad 0 \le n \le N. \tag{36}$$

To prove this, use induction on the output index $n \ge 1$. The residual for the proposed primal output is $(p_i, 0)$, where $i = n \bmod 3$. Its normal tests against the cylinder reduce to the first family in (33). The residual for the proposed dual output is

$$(w_i, (n + 1/8)\varepsilon).$$

The lower-vertex tests are the second family in (33). The remaining apex test is

$$-s + \frac{2(n + 1/8)\varepsilon^2}{r} \le 0$$

by (35). Every vertex is tested, hence the inequalities hold on the whole tetrahedron. The projection characterization

$$u = P_A z \quad \Longleftrightarrow \quad \langle z - u, v - u \rangle \le 0 \quad (v \in A)$$

proves both actual projections. The initial case is by definition, so the induction proves (36). Simultaneously multiplying both sets and both initial points by $a > 0$ multiplies the whole orbit by a, because $P_{aA}(az) = aP_A(z)$.

4.2 Choose the scales and define the Hilbert product

For each integer $j \ge 1$, set

$$a_j = 1/j, \qquad \varepsilon_j = 2^{-j}/8, \qquad N_j = 4^j, \qquad C_j = a_j C^{\varepsilon_j}, \quad D_j = a_j D^{\varepsilon_j}. \tag{37}$$

The horizon check holds uniformly:

$$2(N_j + 1/8)\varepsilon_j^2 = \frac{1}{32} + \frac{1}{256 \cdot 4^j} \le \frac{33}{1024} < \frac{9}{256} < \gamma. \tag{38}$$

Thus block j has the exact scaled prefix through and including time N_j, with third primal coordinate $na_j\varepsilon_j = n/(8j2^j)$.

On the separable real Hilbert space $E = \ell^2(\{1, 2, \ldots\}; E_0)$ define

$$C = \{x \in E : x_j \in C_j \text{ for all } j\}, \qquad D = \{y \in E : y_j \in D_j \text{ for all } j\}.$$

Each block set is nonempty, closed, convex, and contains zero. The product constraint sets are therefore nonempty closed convex subsets of E. Their projections are genuine coordinatewise projections. Indeed, for $z \in E$, let $p_j = P_{C_j} z_j$. The projection inequality at zero gives $\|p_j\| \le \|z_j\|$, so $p = (p_j)$ belongs to E and C. Summing coordinatewise distance minimality against any $w \in C$ proves $p = P_C z$. The same argument applies to D. Therefore every finite iterate of (28) is well defined in E, and its coordinates obey the corresponding block iterations.

Initialize

$$(x_0)_j = (a_j c_0, 0), \qquad (y_0)_j = (a_j d_0, -a_j \varepsilon_j / r). \tag{39}$$

The common planar dual radius is

$$\beta = \|d_i\|_M^2 = \frac{91}{96} - \frac{11\sqrt{91}}{192} > 0.$$

Every point of D_j has norm at most

$$b_j = a_j(\beta + \varepsilon_j^2/r^2)^{1/2},$$

since this bounds every vertex. The elementary estimates $\sum_{j\ge1} j^{-2} \le 2$ and $\sum_{j\ge1} 4^{-j} = 1/3$ yield

$$\sum_{j\ge1} b_j^2 = \beta \sum_{j\ge1} j^{-2} + \frac{1}{64r^2} \sum_{j\ge1} 4^{-j} j^{-2} \le 2\beta + \frac{1}{192r^2} < \infty. \tag{40}$$

Also $\|x_0\|^2 = \sum_{j\ge1} j^{-2} \le 2$, while $\|y_0\|^2 = \sum_{j\ge1} b_j^2$. Hence the initialized pair lies in $C \times D \subset E \times E$.

4.3 Compactness and uniqueness of the KKT pair

The tail estimates are uniform over D. For $J \ge 1$,

$$\sum_{j>J} b_j^2 \le \frac{\beta}{J} + \frac{4^{-J}}{192r^2(J+1)^2} \longrightarrow 0, \tag{41}$$

by an integral bound for $\sum_{j>J} j^{-2}$ and a geometric-series bound for the second term. The first J coordinates form a finite product of compact tetrahedra and have finite nets. Extending these net points by zero in the other coordinates is admissible, because every D_j contains zero. Together with (41), this proves total boundedness of D. The set is closed in the complete space E, so it is norm compact. Every dual iterate, including the initialized point, belongs to D and obeys the bound (40).

Zero is interior to each cylinder C_j in the finite-dimensional space E_0. Each tetrahedron is full dimensional and contains zero with strictly positive barycentric weights

$$0 = \tfrac{1}{2}v^+ + \tfrac{1}{6}v_0^- + \tfrac{1}{6}v_1^- + \tfrac{1}{6}v_2^-.$$

Consequently zero is interior to each D_j as well. Neither product has norm interior: the planar projections of C_j lie in balls of radius $a_j \to 0$, and the entire coordinate radii of D_j are at most $b_j \to 0$. A fixed-radius product-space ball would allow that radius of variation in every coordinate, contradicting these shrinking widths. Only the finite-coordinate interiority is used.

Use the outward normal convention $\eta \in N_A(z)$ if $\langle \eta, w - z\rangle \le 0$ for all $w \in A$. Certainly $(0,0)$ satisfies $-y \in N_C(x)$, $x \in N_D(y)$. Conversely, from any such KKT pair, finite-coordinate competitors give

$$-y_j \in N_{C_j}(x_j), \qquad x_j \in N_{D_j}(y_j).$$

Testing both at zero gives $\langle x_j, y_j\rangle = 0$. The second inclusion then implies $\langle x_j, v\rangle \le 0$ for every $v \in D_j$. A ball around zero inside this finite-dimensional set forces $x_j = 0$. The first inclusion now reads $-y_j \in N_{C_j}(0) = \{0\}$, forcing $y_j = 0$. This holds for every coordinate, proving that the KKT set is exactly $\{(0,0)\}$. The functions ι_C, ι_D are proper lower-semicontinuous convex, so this is exactly the KKT system for (28).

4.4 An active block at every time

For an integer $n \ge 1$, choose the smallest $j \ge 1$ with $n \le 4^j$. Then

$$n \le 4^j \le 4n, \qquad j \le 1 + \log_4 n.$$

For $j > 1$ this follows from minimality, and for $j = 1$ from $n \ge 1$. Block j is still within its proved prefix. Therefore

$$\|x_n\| \ge \frac{n}{8j2^j} \ge \frac{\sqrt{n}}{16j} \ge \frac{\sqrt{n}}{16(1+\log_4 n)}.$$

This proves (29) at every positive integer, without any assumption about any block after its individual horizon. Positive powers dominate logarithms, giving the final conclusion of Theorem 2.

For clarity, this example is not coercive despite its unique KKT pair. Let $z^{(j)}$ have all coordinates zero except the third component 2^j in block j. Then $z^{(j)} \in C$, $\left\|z^{(j)}\right\| = 2^j \to \infty$, and the primal objective satisfies

$$(\iota_C + \sigma_D)(z^{(j)}) = 2^j a_j \varepsilon_j / r = \frac{1}{8rj} \longrightarrow 0,$$

where $\sigma_D(z) = \sup_{y\in D}\langle z, y\rangle$ is the support function. Thus finite-dimensional coercivity mechanisms cannot be imported into this example. Combined with Corollary 3.1, this

proves that $1/2$ is the optimal universal Hilbert-space polynomial growth exponent at the endpoint: no bound $O(n^\alpha)$ with $\alpha < 1/2$ can hold for every KKT-anchored endpoint orbit. This does not assert an attained $\Theta(\sqrt{n})$ lower rate or an optimal logarithmic correction.

4.5 The geometric scaling as a specialization

Corollary 4.1 (Earlier geometric-scale lower bound)**.** *Use the same blocks, product sets and initialization, replacing* (37) *by*

$$a_j = 2^{-j}, \qquad \varepsilon_j = 4^{-j}/8, \qquad N_j = 16^j.$$

Then D is norm compact, the KKT pair is uniquely $(0,0)$, and

$$\|x_{16^j}\| \geq 2^j/8 \quad (j \geq 1), \qquad \|x_n\| \geq n^{1/4}/64 \quad (n \geq 1),$$

while

$$\|x_0\|^2 = \frac{1}{3}, \qquad \|y_0\|^2 = \frac{\beta}{3} + \frac{1}{4032r^2}, \qquad \|y_n\|^2 \leq \frac{\beta}{3} + \frac{1}{4032r^2} \quad (n \geq 0).$$

Proof. The common block proof applies since $2(N_j + 1/8)\varepsilon_j^2 = 1/32 + 1/(256 \cdot 16^j) \leq 9/256 < \gamma$. Here $\sum a_j^2 = 1/3$ and $\sum a_j^2 \varepsilon_j^2 = 1/4032$, which give the displayed norm sums and the summable coordinate radii used in the projection and compactness proofs. The finite-coordinate proof of KKT uniqueness is unchanged. At N_j the third primal coordinate is $N_j a_j \varepsilon_j = 2^j/8$. For the least $j \geq 1$ with $n \leq 16^j$, one has $16^j \leq 16n$ and

$$\|x_n\| \geq \frac{n}{8 \cdot 8^j} \geq \frac{n^{1/4}}{64}.$$

Thus no duplicate block construction or post-horizon assumption is needed. For the points $z^{(j)}$ used above, this scaling gives $J(z^{(j)}) = 4^{-j}/(8r) \to 0$ while $\left\|z^{(j)}\right\| = 2^j$, again excluding norm coercivity. ∎

5 Boundedness from a coercive primal objective

The lower example separates uniqueness from coercivity. The following criterion shows that, once the dual domain is bounded, norm coercivity of the whole primal objective excludes escape. It does not rely on the endpoint increment certificate and holds at all positive steps. A more explicit interior-ball criterion and boundedness of each individual drifting block follow below.

Theorem 3. *Let X, Y be real Hilbert spaces, $L : X \to Y$ bounded and linear, and f, h proper lower-semicontinuous convex functions on X, Y, respectively. Suppose* (2) *holds,* $\operatorname{dom} h$ *is bounded, and*

$$J(x) = f(x) + h^*(Lx) \tag{42}$$

is norm coercive: $J(x) \to +\infty$ whenever $\|x\| \to \infty$. For arbitrary initial points, all $\tau, \sigma > 0$, and any $\theta \geq 0$, the iteration (1) has bounded primal and dual sequences. No step-product restriction is imposed, and no convergence conclusion is asserted.

In normalized coordinates where $f(0) = \min f = 0$, $h(0) = \min h = 0$, and $\operatorname{dom} h \subset \{y : \|y\| \leq M\}$ with $M > 0$, put

$$a = \|L\|, \qquad A_c = 4\tau a M, \qquad B_c = 4\tau\theta a M, \qquad \Gamma_c = \frac{8\tau M^2}{\sigma} + A_c^2 + B_c^2/2. \tag{43}$$

Writing $d_n = x_n - x_{n-1}$ and $\mathcal{E}_n = \|x_n\|^2 + \frac{1}{2}\|d_n\|^2$, one has

$$\mathcal{E}_{n+1} \leq \mathcal{E}_n - \tfrac{1}{4}\|d_{n+1}\|^2 - 2\tau J(x_{n+1}) + \Gamma_c \qquad (n \geq 2). \tag{44}$$

Choose $Q \geq 0$ such that $2\tau J(x) \geq \Gamma_c$ whenever $\|x\| \geq Q$. Then

$$\mathcal{E}_n \leq \max\{\mathcal{E}_2, Q^2 + 2\Gamma_c\} \quad (n \geq 2), \qquad \|y_n\| \leq M \quad (n \geq 1). \tag{45}$$

No bound on increments is assumed.

Proof. *KKT normalization.* Fix a KKT pair (x_*, y_*). Put $u = x - x_*$, $v = y - y_*$ and

$$\begin{aligned} F(u) &= f(x_* + u) - f(x_*) + \langle L^* y_*, u\rangle, \\ G(v) &= h(y_* + v) - h(y_*) - \langle L x_*, v\rangle. \end{aligned} \tag{46}$$

The KKT inequalities show that F, G are nonnegative and vanish at zero. They remain proper, lower semicontinuous, and convex, while $\operatorname{dom} G = \operatorname{dom} h - y_*$ remains bounded. Substitution in the two proximal optimality inclusions gives

$$\begin{aligned} (u_{n-1} - u_n)/\tau - L^* v_{n-1} &\in \partial F(u_n), \\ (v_{n-1} - v_n)/\sigma + L(u_n + \theta(u_n - u_{n-1})) &\in \partial G(v_n). \end{aligned} \tag{47}$$

Thus the translated orbit is exactly the same algorithm for F, G. The conjugate is

$$G^*(z) = h^*(z + Lx_*) - \langle z + Lx_*, y_*\rangle + h(y_*). \tag{48}$$

Fenchel equality at the KKT pair gives $h(y_*) + h^*(Lx_*) = \langle Lx_*, y_*\rangle$, so

$$F(u) + G^*(Lu) = J(x_* + u) - J(x_*). \tag{49}$$

In particular, x_* minimizes J and coercivity is preserved by this normalization. It suffices to prove the quantitative normalized claim. In the remainder of the proof use the names f, h, x_n, y_n for these normalized data and iterates.

A dual support estimate. The normalized conjugate is finite everywhere and satisfies

$$0 \leq h^*(z) \leq M\|z\|, \qquad |h^*(z) - h^*(w)| \leq M\|z - w\|. \tag{50}$$

Indeed, the zero dual point gives the lower bound, and $h(v) \geq 0$, $\|v\| \leq M$ on its domain give the upper bound. For Lipschitz continuity, compare $\langle z, v\rangle - h(v)$ and $\langle w, v\rangle - h(v)$

for each domain point and take suprema in both directions. No maximizing dual point is needed.

For $n \geq 2$, both y_{n-1} and y_n have norm at most M. The actual dual inclusion reads

$$Lx_n + e_n \in \partial h(y_n), \qquad e_n = (y_{n-1} - y_n)/\sigma + \theta L d_n, \quad \|e_n\| \leq 2M/\sigma + \theta a \|d_n\| . \tag{51}$$

For every $v \in \operatorname{dom} h$, its subgradient inequality gives

$$\begin{aligned} \langle Lx_n, y_n \rangle &\geq \langle Lx_n, v \rangle - h(v) + h(y_n) + \langle e_n, v - y_n \rangle \\ &\geq \langle Lx_n, v \rangle - h(v) - 2M \|e_n\| . \end{aligned} \tag{52}$$

Taking the supremum over v proves

$$\langle Lx_n, y_n \rangle \geq h^*(Lx_n) - \frac{4M^2}{\sigma} - 2M\theta a \|d_n\| . \tag{53}$$

Keep the primal function value. Testing the next primal proximal inclusion at zero yields

$$\|x_{n+1}\|^2 \leq \|x_n\|^2 - \|d_{n+1}\|^2 - 2\tau f(x_{n+1}) - 2\tau \langle Lx_{n+1}, y_n \rangle . \tag{54}$$

Replace Lx_{n+1} by $Lx_n + Ld_{n+1}$, bounding the added pairing by $aM \|d_{n+1}\|$, and use (53). By (50), replacing $h^*(Lx_n)$ by $h^*(Lx_{n+1})$ costs at most another $Ma \|d_{n+1}\|$. Consequently,

$$\begin{aligned} \|x_{n+1}\|^2 \leq \|x_n\|^2 - \|d_{n+1}\|^2 + A_c \|d_{n+1}\| + B_c \|d_n\| \\ - 2\tau J(x_{n+1}) + 8\tau M^2/\sigma. \end{aligned} \tag{55}$$

The objective is finite at these positive-index iterates: f is finite at its proximal outputs and h^* is finite everywhere. Add $\frac{1}{2}\|d_{n+1}\|^2$ and write the current squared norm as $\mathcal{E}_n - \frac{1}{2}\|d_n\|^2$. The elementary inequalities

$$-\tfrac{1}{2}s^2 + A_c s \leq -\tfrac{1}{4}s^2 + A_c^2, \qquad -\tfrac{1}{2}t^2 + B_c t \leq B_c^2/2 \tag{56}$$

then prove (44).

An invariant bound from implicit descent. The normalized J is nonnegative and $J(0) = 0$. Coercivity supplies Q as in the statement. Set $R_c = Q^2 + 2\Gamma_c$. If $\mathcal{E}_{n+1} > R_c$, then either $\|x_{n+1}\| > Q$ or $\|d_{n+1}\|^2 > 4\Gamma_c$. In the first case the objective term in (44) absorbs Γ_c; in the second case the squared-increment term does so, using $J \geq 0$. Thus $\mathcal{E}_{n+1} \leq \mathcal{E}_n$ whenever $\mathcal{E}_{n+1} > R_c$. There can be no first crossing above $\max\{\mathcal{E}_2, R_c\}$, which proves (45) by induction. The remaining initial points are finite, and all positive-index dual points have norm at most M. Translation back finishes the proof. ∎

5.1 Bounded primal minimizers in finite dimensions

Corollary 5.1. *Let X, Y be finite-dimensional real Hilbert spaces and $L : X \to Y$ linear. Let f, h be proper lower semicontinuous convex functions. Suppose the KKT set is*

nonempty and $\operatorname{dom} h$ *is bounded. If the primal minimizer set of* $J = f + h^* \circ L$ *is bounded, every initialized orbit of* (1) *is bounded for all* $\tau, \sigma > 0$ *and* $\theta \geq 0$*. In particular, a unique KKT pair together with a bounded dual domain suffices. Injectivity of* L *and interiority of the dual domain are not required in this corollary.*

Proof. Normalize at a KKT pair as in (46)–(49). The normalized J is proper, lower semicontinuous, and convex, since h^* is finite and Lipschitz. It is nonnegative and vanishes at zero. Suppose a finite sublevel $\{J \leq \alpha\}$ were unbounded. Then $\alpha \geq 0$, and there are z_j in that sublevel with $\|z_j\| \to \infty$. A subsequence satisfies $z_j / \|z_j\| \to d$ with $\|d\| = 1$, by finite-dimensional compactness of the unit sphere. For each fixed $t \geq 0$ and all sufficiently large j, convexity gives

$$J\left(\frac{tz_j}{\|z_j\|}\right) \leq \frac{t\alpha}{\|z_j\|}. \tag{57}$$

Lower semicontinuity implies $J(td) = 0$. The resulting unbounded minimizing ray contradicts boundedness of the minimizer set. Every finite sublevel is therefore bounded, which gives norm coercivity. Theorem 3 applies. If $X = \{0\}$, there is no unbounded sublevel and the conclusion is immediate as well.

For the last assertion, fix the unique KKT pair (x_*, y_*) and let x be any primal minimizer. The supporting inequalities at x_* for f and at Lx_* for h^* are

$$\begin{aligned} f(x) - f(x_*) + \langle L^* y_*, x - x_* \rangle &\geq 0, \\ h^*(Lx) - h^*(Lx_*) - \langle y_*, Lx - Lx_* \rangle &\geq 0. \end{aligned} \tag{58}$$

Their sum is $J(x) - J(x_*) = 0$, so both are equalities. An affine supporting function touching at x_* also touches at x in the first inequality, and similarly at Lx in the second. Thus $-L^* y_* \in \partial f(x)$ and $y_* \in \partial h^*(Lx)$, the latter equivalent to $Lx \in \partial h(y_*)$. Hence (x, y_*) is KKT and uniqueness forces $x = x_*$. The primal minimizer set is a singleton, so the first part applies. ∎

The finite-dimensional conclusion with a unique KKT pair and bounded dual domain contrasts with Theorem 2. It does not settle the unrestricted finite-dimensional endpoint problem when these additional conditions are absent.

6 A quantitative restoring ball theorem

The following hypotheses imply the coercivity criterion of Section 5. Indeed, for nonnegative f and a dual testing ball of radius δ on which $h \leq H$,

$$J(x) = f(x) + h^*(Lx) \geq \delta \|Lx\| - H \geq \delta \ell \|x\| - H. \tag{59}$$

Thus qualitative boundedness follows from Theorem 3. The direct proof below retains the explicit constants of the more specialized restoring-ball estimate. The product set in Theorem 2 has no such norm-interior ball.

Theorem 4. *Let X, Y be real Hilbert spaces and $L : X \to Y$ bounded and linear. Let $f : X \to (-\infty, +\infty]$ and $h : Y \to (-\infty, +\infty]$ be proper lower-semicontinuous convex functions with $f(0) = \min f = 0$ and $h(0) = \min h = 0$. Suppose there exist $\ell, M, \delta > 0$ and $H \geq 0$ such that*

$$\begin{aligned} &\|Lx\| \geq \ell \|x\| && (x \in X), \\ &\operatorname{dom} h \subset \{y : \|y\| \leq M\}, \qquad \{y : \|y\| \leq \delta\} \subset \operatorname{dom} h, && \\ &h(y) \leq H && (\|y\| \leq \delta). \end{aligned} \tag{60}$$

For arbitrary initialization, positive τ, σ, and $\theta \geq 0$, both sequences in (1) are bounded, without any restriction on $\tau\sigma \|L\|^2$. More precisely, set

$$\begin{aligned} a = \|L\|, \qquad A = 2\tau a M, \qquad B = 2\tau\theta a(M + \delta), \\ C_0 = 2\tau \left[H + \frac{2M(M+\delta)}{\sigma} \right], \qquad \Gamma = C_0 + A^2/2 + B^2, \\ k = 2\tau\delta\ell, \qquad R_0 = (2\Gamma/k)^2 + 2\Gamma. \end{aligned} \tag{61}$$

With $d_n = x_n - x_{n-1}$ and $\mathcal{E}_n = \|x_n\|^2 + \frac{1}{2}\|d_n\|^2$, one has

$$\begin{aligned} &\mathcal{E}_{n+1} \leq \mathcal{E}_n - \tfrac{1}{4}\|d_n\|^2 - k\|x_n\| + \Gamma && (n \geq 2), \\ &\mathcal{E}_n \leq \max\{\mathcal{E}_2, R_0 + \Gamma\} && (n \geq 2), \\ &\|y_n\| \leq M && (n \geq 1). \end{aligned} \tag{62}$$

No increment bound is assumed, and no convergence conclusion is asserted.

Proof. Every positive-index y_n belongs to $\operatorname{dom} h$, so $\|y_n\| \leq M$. For $n \geq 2$, the dual optimality inclusion is

$$Lx_n + e_n \in \partial h(y_n), \qquad e_n = \frac{y_{n-1} - y_n}{\sigma} + \theta L d_n, \quad \|e_n\| \leq \frac{2M}{\sigma} + \theta a \|d_n\|. \tag{63}$$

If $Lx_n \neq 0$, choose $v_n = \delta Lx_n / \|Lx_n\|$; otherwise choose $v_n = 0$. Then $\|v_n\| \leq \delta$, $h(v_n) \leq H$, and $\langle Lx_n, v_n \rangle = \delta \|Lx_n\|$. The subgradient inequality at this actual feasible point gives

$$\begin{aligned} \langle Lx_n, y_n \rangle &\geq \delta \|Lx_n\| - H + \langle e_n, v_n - y_n \rangle \\ &\geq \delta \|Lx_n\| - H - (M + \delta)\left(\frac{2M}{\sigma} + \theta a \|d_n\| \right). \end{aligned} \tag{64}$$

Here $h(y_n) \geq 0$ has been used. The finite constant H is the additional term for general dual functions; it vanishes for an indicator.

For the primal update from n to $n+1$, the optimality inclusion tested at the minimizer zero yields

$$\langle x_n - x_{n+1} - \tau L^* y_n, x_{n+1} \rangle \geq \tau f(x_{n+1}) \geq 0. \tag{65}$$

The identity $2\langle x_n - x_{n+1}, x_{n+1} \rangle = \|x_n\|^2 - \|x_{n+1}\|^2 - \|d_{n+1}\|^2$ and Cauchy–Schwarz therefore give

$$\begin{aligned} \|x_{n+1}\|^2 &\leq \|x_n\|^2 - \|d_{n+1}\|^2 - 2\tau \langle Lx_{n+1}, y_n \rangle \\ &\leq \|x_n\|^2 - \|d_{n+1}\|^2 - 2\tau \langle Lx_n, y_n \rangle + 2\tau a M \|d_{n+1}\|. \end{aligned} \tag{66}$$

Insert (64) and use $\|Lx_n\| \geq \ell \|x_n\|$ to obtain

$$\|x_{n+1}\|^2 \leq \|x_n\|^2 - \|d_{n+1}\|^2 + A\|d_{n+1}\| + B\|d_n\| - k\|x_n\| + C_0. \tag{67}$$

Adding half of the next squared increment and subtracting half of the current one on the right converts this to

$$\begin{aligned} \mathcal{E}_{n+1} \leq \mathcal{E}_n &- \tfrac{1}{2}\|d_{n+1}\|^2 + A\|d_{n+1}\| \\ &- \tfrac{1}{2}\|d_n\|^2 + B\|d_n\| - k\|x_n\| + C_0. \end{aligned} \tag{68}$$

Completing squares, for nonnegative s, t,

$$-\tfrac{1}{2}s^2 + As \leq A^2/2, \qquad -\tfrac{1}{2}t^2 + Bt \leq -\tfrac{1}{4}t^2 + B^2. \tag{69}$$

This proves the first inequality in (62).

For completeness, it supplies an invariant bound without any previously known control on d_n. If $\mathcal{E}_n > R_0$, then at least one of $\|x_n\| > 2\Gamma/k$ or $\|d_n\|^2 > 4\Gamma$ holds; otherwise the definition of $\mathcal{E}_n$ would give $\mathcal{E}_n \leq R_0$. In either case the first inequality in (62) implies $\mathcal{E}_{n+1} \leq \mathcal{E}_n$. If instead $\mathcal{E}_n \leq R_0$, the same inequality gives $\mathcal{E}_{n+1} \leq R_0 + \Gamma$. Induction from $n = 2$ proves the second inequality. All proximal mappings are everywhere defined, so x_0, x_1, x_2, y_0 and $\mathcal{E}_2$ are finite. Together with the positive-index dual bound, this bounds both full sequences. ∎

6.1 Finite dimensional KKT anchoring

Corollary 6.1. *Let X, Y be finite-dimensional real Hilbert spaces, $L : X \to Y$ injective, and f, h proper lower-semicontinuous convex functions. Suppose* $\operatorname{dom} h$ *is bounded and there is a KKT pair (x_*, y_*) such that*

$$-L^*y_* \in \partial f(x_*), \qquad Lx_* \in \partial h(y_*), \qquad y_* \in \operatorname{int}(\operatorname{dom} h). \tag{70}$$

Then every initialized orbit of (1) *is bounded, for every $\tau, \sigma > 0$ and $\theta \geq 0$.*

Proof. Normalize at the KKT pair using (46)–(47). The translated functions F, G are nonnegative and vanish at zero; $\operatorname{dom} G = \operatorname{dom} h - y_*$ is bounded and contains zero in its interior. The translated iterates follow exactly the same algorithm for F, G.

If $X = \{0\}$, the primal sequence is constant and the dual outputs belong to a bounded domain, which proves the conclusion directly. Otherwise, compactness of the finite-dimensional unit sphere and injectivity of L give $\ell = \min_{\|u\|=1} \|Lu\| > 0$. Choose $M > 0$ bounding $\operatorname{dom} G$. A sufficiently small coordinate cube centered at zero has all its finitely many vertices in $\operatorname{dom} G$ and contains a closed ball of some radius $\delta > 0$. Let H be the maximum of G at these vertices. It is finite and nonnegative. Convexity implies $G(v) \leq H$ throughout the cube, hence throughout this ball. This supplies every hypothesis of Theorem 4 for F, G. Its conclusion bounds u_n, v_n; translation back bounds x_n, y_n. ∎

6.2 Bounded blocks and nonuniform transients

Proposition 6.1. Let C, D be nonempty closed convex sets in a real Hilbert space, with $0 \in C$ and

$$\{y : \|y\| \le \delta\} \subset D \subset \{y : \|y\| \le M\} \quad \text{for some } \delta, M > 0. \tag{71}$$

For arbitrary initial points, $\tau, \sigma > 0$, and $\theta \ge 0$, the iteration (1) with $f = \iota_C$, $h = \iota_D$, and $L = I$ has bounded primal and dual sequences. With $\mathcal{E}_n = \|x_n\|^2 + \frac{1}{2}\|x_n - x_{n-1}\|^2$, put

$$\Gamma = \frac{4\tau M(M+\delta)}{\sigma} + 2\tau^2 M^2 + 4\tau^2\theta^2(M+\delta)^2, \qquad k = 2\tau\delta. \tag{72}$$

Then

$$\mathcal{E}_n \le \max\{\mathcal{E}_2, (2\Gamma/k)^2 + 3\Gamma\} \quad (n \ge 2), \qquad \|y_n\| \le M \quad (n \ge 1). \tag{73}$$

Proof. Apply Theorem 4 with $f = \iota_C$, $h = \iota_D$, $L = I$, $a = \ell = 1$, and $H = 0$. Its hypotheses follow from (71) and $0 \in C$. Its constants simplify exactly to (72), and $R_0 + \Gamma = (2\Gamma/k)^2 + 3\Gamma$.

∎

For every fixed $\varepsilon > 0$, the cylinder C^ε contains zero and the compact full-dimensional tetrahedron D^ε contains a closed ball of positive radius. Thus every individual block, every scaled block, and every fixed finite product of blocks has bounded orbits for all positive steps and $\theta \ge 0$. Nevertheless the unscaled geometric family $\varepsilon_j = 4^{-j}/8$, $N_j = 16^j$ has uniformly bounded initial norms and actual transients

$$\left\|x_{N_j}\right\| \ge N_j\varepsilon_j = 4^j/8 \longrightarrow \infty.$$

Each such three-dimensional problem has unique KKT pair $(0, 0)$. There is therefore no uniform orbit bound across this family of thinning tetrahedra and bounded initial pairs, although each orbit is bounded. In the Hilbert product identity coupling remains bounded below; the missing restoring-ball hypothesis is the positive-radius norm interior of D. This explains how infinitely many bounded coordinate orbits can produce the unbounded global orbit of Theorem 2.

7 Structural obstructions to escape

7.1 Periodic affine itineraries

The term "periodic affine itinerary" refers to repetition of affine update formulas, not to periodicity of the positions.

Lemma 1 (Sublinear affine recurrences are bounded)**.** *Let $w_{k+1} = Aw_k + b$ in a finite-dimensional real vector space. If $\|w_k\| = o(k)$, then (w_k) is bounded.*

Proof. Summing the recurrence yields

$$(I - A)\left(\frac{1}{k}\sum_{j=0}^{k-1} w_j\right) = b - \frac{w_k - w_0}{k} \longrightarrow b.$$

The range of $I - A$ is closed in finite dimensions. Thus $b \in \operatorname{ran}(I - A)$, so there is w_* with $(I - A)w_* = b$. Set $v_k = w_k - w_* = A^k v_0 = o(k)$.

Complexify and decompose v_0 into generalized eigenspaces. Every spectral projection is bounded. For a nonzero eigenvalue λ, a nonzero component of Jordan height q evolves as λ^k times a vector-valued polynomial in k of degree $q-1$ with nonzero leading coefficient. A component with $|\lambda| > 1$ cannot satisfy $o(k)$. A component with $|\lambda| = 1$ and height at least two cannot satisfy $o(k)$ either; height one is bounded. If $|\lambda| < 1$, every polynomial-times-geometric component is bounded and tends to zero. Components at zero eigenvalues vanish after finitely many steps. All remaining components are bounded, which proves the claim. Only the generalized eigenspaces excited by v_0 matter; A need not be power bounded on the full ambient space. ∎

Theorem 5 (Periodic affine itinerary bound)**.** *Assume X, Y are finite-dimensional,* (2) *holds, and the orbit of* (1) *has bounded increments. Suppose there exist integers $n_0 \geq 0$, $m \geq 1$ and fixed affine maps $T_0, \ldots, T_{m-1}$ on $X \times Y$ such that, with $z_n = (x_n, y_n)$,*

$$z_{n+1} = T_j(z_n) \quad \textit{whenever } n \geq n_0, \quad j \equiv n - n_0 \pmod{m}, \quad 0 \leq j < m.$$

Then the whole orbit is bounded. At (5)*, the separate bounded-increment assumption is automatic.*

Proof. Theorem 1 gives $\|z_n\| = O(\sqrt{n})$ in any fixed product norm, since the normalizations are equivalent and fixed translations do not affect growth. Compose one full period:

$$T = T_{m-1} \circ \cdots \circ T_0, \qquad T(z) = Az + b.$$

Then $w_k = z_{n_0+km}$ obeys $w_{k+1} = Aw_k + b$ and $\|w_k\| = O(\sqrt{k}) = o(k)$. Lemma 1 bounds this subsequence. Each other residue class is a fixed finite affine composition of it and is bounded as well. Adding the finite prefix completes the proof. Proposition 3.1 supplies the endpoint case. ∎

For a nonempty closed polyhedron C in a finite-dimensional space, its projection is one of finitely many affine formulas. Indeed, if the projected point lies in the relative interior of a face F, the projection residual is perpendicular to the direction space of $\operatorname{aff} F$. The projection therefore equals the orthogonal projection onto that affine hull. There are only finitely many faces, and the corresponding normal-cone conditions specify the regions where each formula applies. For $f = \iota_C$ and $h = \iota_D$, a complete step of (1) is consequently one of finitely many affine maps. Theorem 5 applies if the formulas selected

by the actual orbit eventually repeat periodically. It also applies to other piecewise-affine proximal maps under the same itinerary hypothesis.

A finite list of formulas does not imply eventual periodic switching. The theorem excludes endpoint escape along a fixed affine piece or a forever-repeating finite sequence of pieces, but it does not treat an infinitely varying nonperiodic itinerary. Nor does boundedness here assert convergence to a KKT point or to a periodic orbit.

7.2 An obstruction for monotone graphs

The signed-storage proof of Theorem 1 uses convex function values. The following weaker conclusion needs only monotonicity and so has a broader operator scope. It is useful when subdifferential structure is absent.

Let $\mathcal{A} : X \rightrightarrows X$ and $\mathcal{B} : Y \rightrightarrows Y$ be monotone relations. Assume an actual orbit and a graph anchor satisfy

$$\begin{aligned} p_n &= -u_n/\tau - L^* y_n \in \mathcal{A} x_{n+1}, & u_n &= x_{n+1} - x_n, \\ q_n &= -v_n/\sigma + L x_{n+1} + \theta L u_n \in \mathcal{B} y_{n+1}, & v_n &= y_{n+1} - y_n, \\ -L^* y_* &\in \mathcal{A} x_*, & L x_* &\in \mathcal{B} y_*. \end{aligned} \tag{74}$$

In this section u_n, v_n denote raw increments, not the normalized positions of Section 2. We condition on the orbit's existence; no resolvent existence or uniqueness is inferred for arbitrary monotone relations. For $\mathcal{A} = \partial f$, $\mathcal{B} = \partial h$, (74) is exactly (1) with its natural subgradients.

Theorem 6 (No nonzero limiting average velocity)**.** *Let $\tau, \sigma > 0$ and $\theta \in \mathbb{R}$ be arbitrary. Under* (74)*, suppose u_n, v_n are uniformly bounded and*

$$x_n/n \longrightarrow a, \qquad y_n/n \longrightarrow b \quad \text{in norm.}$$

Then $a = b = 0$.

Proof. Put $c = \langle La, b \rangle$. Separate monotonicity against the graph anchor gives

$$F_n = \langle x_{n+1} - x_*, p_n + L^* y_* \rangle \geq 0, \qquad H_n = \langle y_{n+1} - y_*, q_n - L x_* \rangle \geq 0.$$

The normalized position limits and bounded increments imply

$$F_n/n^2 \longrightarrow -c, \qquad H_n/n^2 \longrightarrow c.$$

Therefore $c = 0$. Before summing, cancel the cross terms exactly:

$$\begin{aligned} M_n := F_n + H_n = &- \langle x_{n+1} - x_*, u_n \rangle / \tau - \langle y_{n+1} - y_*, v_n \rangle / \sigma \\ &+ \langle L(x_{n+1} - x_*), v_n \rangle + \theta \langle L u_n, y_{n+1} - y_* \rangle \geq 0. \end{aligned} \tag{75}$$

Write $x_n = na + r_n$ and $y_n = nb + s_n$, where $\|r_n\|, \|s_n\| = o(n)$. Discrete summation by parts gives, in norm,

$$\sum_{n=0}^{T-1}(n+1)(s_{n+1} - s_n) = Ts_T - \sum_{n=0}^{T-1} s_n = o(T^2), \tag{76}$$

and the same identity holds for r_n. Moreover,

$$\sum_{n=0}^{T-1} \|r_{n+1}\| = o(T^2), \qquad \sum_{n=0}^{T-1} \|s_{n+1}\| = o(T^2). \tag{77}$$

For example, for each $\varepsilon > 0$ the tail satisfies $\|r_n\| \le \varepsilon n$, and its sum is at most $\varepsilon T(T+1)/2$; the omitted initial sum is fixed. Division by T^2 and then $\varepsilon \downarrow 0$ prove the claim.

Using $v_n = b + s_{n+1} - s_n$, we have

$$\sum_{n=0}^{T-1} \langle Lx_{n+1}, v_n\rangle = c\,\frac{T(T+1)}{2} + \left\langle La, \sum_{n=0}^{T-1}(n+1)(s_{n+1} - s_n)\right\rangle + \sum_{n=0}^{T-1} \langle Lr_{n+1}, v_n\rangle = c\,\frac{T(T+1)}{2} + o(T^2).$$

Here the last error uses bounded v_n and (77). Interchanging the roles of the expansions and using bounded u_n likewise gives

$$\sum_{n=0}^{T-1} \langle Lu_n, y_{n+1}\rangle = c\,\frac{T(T+1)}{2} + o(T^2).$$

The fixed-anchor contributions in the cross sums telescope to $O(T)$.

Finally, polarization gives the exact identity

$$\sum_{n=0}^{T-1} \langle x_{n+1} - x_*, u_n\rangle = \frac{1}{2}\left(\|x_T - x_*\|^2 - \|x_0 - x_*\|^2 + \sum_{n=0}^{T-1}\|u_n\|^2\right),$$

and its dual analogue. The square sums are $O(T)$. Summing (75), dividing by T^2, and taking the limit yields

$$0 \le -\frac{\|a\|^2}{2\tau} - \frac{\|b\|^2}{2\sigma} + \frac{1+\theta}{2}c = -\frac{\|a\|^2}{2\tau} - \frac{\|b\|^2}{2\sigma}.$$

Positive steps force $a = b = 0$. The two cross sums were evaluated separately, so no sign condition on θ was used. ∎

Corollary 7.1 (Indexed affine approximation and affine-periodic tails)**.** *Under* (74)*, an indexed expansion*

$$(x_n, y_n) = (\bar{x}, \bar{y}) + n(a, b) + O(1)$$

with a uniformly norm-bounded error forces $a = b = 0$ and therefore bounds this orbit. Also, if for finitely many phases $0 \le i < m$,

$$x_{N+km+i} = ka + X_i + o(1), \qquad y_{N+km+i} = kb + Y_i + o(1) \quad (k \to \infty) \tag{78}$$

in norm, then $a = b = 0$. In particular, an exact tail satisfying $x_{n+m} = x_n + a$ and $y_{n+m} = y_n + b$ has zero drift.

Proof. A bounded indexed error implies bounded increments and the required average-velocity limits; apply Theorem 6. In (78), subtract na/m and nb/m. For $n = N+km+i$, the remaining terms are $X_i - (N+i)a/m + o(1)$ and $Y_i - (N+i)b/m + o(1)$, uniformly bounded over the finite set of phases. Thus the indexed result applies with velocities $a/m, b/m$. Exact affine-periodicity is the special case with zero errors and the appropriate phase vectors. ∎

The indexed approximation condition concerns the position at each prescribed n. Mere bounded distance from a geometric ray, with no relation between the ray parameter and n, is a different statement and is not used. In the convex setting with bounded increments, Theorem 1 is stronger: it proves the normalized limits are zero without assuming that those limits exist. Theorem 6 remains useful because it applies to separate monotone graphs without convex potentials.

8 Scope and remaining endpoint questions

The assumptions enter at distinct stages. Bounded increments, convex potentials, and one KKT pair give the square-root bound in arbitrary Hilbert spaces for every fixed real θ and all positive steps. At (5), the exact certificate supplies bounded increments without a KKT pair; the anchor is needed for the improved position estimate. An ambient finite-dimensional hypothesis is used in the affine recurrence lemma and its periodic-itinerary consequence. The monotone-graph theorem requires a graph anchor, bounded increments, and existing norm limits, with no step or extrapolation restriction.

Theorem 2 uses one explicit block construction; Corollary 4.1 retains the earlier $n^{1/4}$ scaling as a specialization, rather than a second construction. Its single orbit grows faster than n^α for every $\alpha < 1/2$, while Corollary 3.1 bounds every KKT-anchored endpoint orbit by order $\sqrt{n}$. The smallest $p \ge 0$ for which every KKT-anchored Hilbert-space endpoint orbit is $O(n^p)$, with constants depending on the orbit, is therefore $1/2$. This does not produce an attained $\Theta(\sqrt{n})$ orbit, prove the optimality of the logarithmic factor, or settle whether every individual orbit is $o(\sqrt{n})$. The lower example is infinite-dimensional and uses only long finite drifting prefixes in each block. Theorems 3 and 4 give separate sufficient boundedness conditions, while Corollary 5.1 uses finite-dimensional compactness to turn bounded minimizers into coercivity. A unique KKT pair without that compactness

does not do so. No unrestricted nonseparable finite-dimensional endpoint boundedness, general polyhedral boundedness, or endpoint convergence conclusion follows.

Normalized-iterate limits for nonexpansive maps have a classical history; see Pazy [2]. For a nonexpansive map with a fixed point in a genuine norm, the elementary estimate $\|T^n z - T^n z_*\| \leq \|z - z_*\|$ already gives boundedness. In primal-dual hybrid gradient analysis, Applegate, Díaz, Lu, and Lubin [3], Proposition 2, use a positive-definite metric under a strict step-product restriction; their Algorithm 1 uses unit extrapolation. The arguments here do not assume such a metric and allow all positive steps and real extrapolation under their displayed orbit hypotheses. The KKT anchor also distinguishes these statements from drift associated with infeasible optimization problems. This comparison concerns hypotheses, not priority or a contradiction with those results.

Reproducibility

The accompanying archive supplies all LaTeX modules, the unchanged reference style, a deterministic build script, exact arithmetic checkers, coefficient data and hash manifests. The endpoint checker reconstructs the finite certificate, including support, value cancellations, kernel bases and positive complementary blocks. The lower-growth checker verifies the common planar block and the square-root rescaling; the retained geometric-scale checkers verify the earlier specialization independently. None uses a numerical optimizer or a floating-point positivity test. All-index inequalities, Hilbert-product well-definedness and compactness, KKT uniqueness, the growth storage, and the boundedness and monotone-graph results have analytic proofs here; finite checks are not substitutes for these proofs. The public result map records the location or subsumption of every mathematical outcome consolidated from reports 32 and 40. The separately identified report 30 is an unchanged prior prerequisite.

References

[1] A. Chambolle and T. Pock, A first-order primal-dual algorithm for convex problems with applications to imaging, *Journal of Mathematical Imaging and Vision* 40 (2011), 120–145. https://doi.org/10.1007/s10851-010-0251-1.

[2] A. Pazy, Asymptotic behavior of contractions in Hilbert space, *Israel Journal of Mathematics* 9 (1971), 235–240. https://doi.org/10.1007/BF02771588.

[3] D. Applegate, M. Díaz, H. Lu, and M. Lubin, Infeasibility detection with primal-dual hybrid gradient for large-scale linear programming, arXiv:2102.04592v1 (2021). https://arxiv.org/abs/2102.04592v1.

[4] *The sharp universal convergence threshold for Chambolle–Pock at one eighth extrapolation*, companion mathematical report 30 (2026). Unchanged prior prerequisite PDF, source, exact coefficient file and verification programs are included in the dependency directory of the present source archive.

[5] *Primal escape at the critical Chambolle–Pock endpoint with a unique KKT point*, companion mathematical report 32 (2026). Source of the earlier geometric-scale construction and sufficient boundedness results, integrated and proved in this manuscript; not an additional theorem premise.

[6] M. Upadhyaya, The Chambolle–Pock method converges weakly with $0 < \theta \leq 1/2$ and $\tau\sigma \|L\|^2 < 4\theta(2-\theta)/(1-2\theta+9\theta^2-4\theta^3)$, arXiv:2604.06423v2 (20 August 2026). `https://arxiv.org/abs/2604.06423v2`.

[7] S. Banert, M. Upadhyaya, and P. Giselsson, The Chambolle–Pock method converges weakly with $\theta > 1/2$ and $\tau\sigma \|L\|^2 < 4/(1+2\theta)$, *Optimization Letters* 20 (2026), 503–520. `https://doi.org/10.1007/s11590-025-02250-0`.